An experimental study to evaluate lncRNA SNHG1 gene expression levels in cervical cancer patients.

Introduction:

Cervical cancer is often observed in females and it is caused by the infection of the Human Papilloma Virus(HPV) due to its mutation in gene expression persuading factor. Chlamydia trachomatis, smoking, drinking, unhygienic practices of intimate areas are root causes of the disease. The primary cause of CC by Human papillomavirus infection and non-coding RNA and microRNA's are the key players observed in molecular pathogenesis. However early diagnosis and treatment, vaccination prevents major risks of CC.

Materials and Methods:

A combination of human epidermal cells (HaCaT) and ATCC bank infected cancer cell lines. The medium consists of a basic DMEM medium, 10 % fetal bovine serum (FBS), and 1% penicillin-streptomycin solution. The cells were solidified, incubated at 37 ° C until well grown, and the cells were transfected with lncRNA SNHG1 siRNA, miR195mimics, miR195 inhibitors and NC vectors using the Lipofectamine 2000 transfection kit. A new cell growth formed after 8 hrs of incubation.

Results:

Quantification of SNHG1 and miR195 lncRNAs by RTPCR in 54 (ordinary tissues) and 72 (numerous tissues infected with cervical cancer). SNHG1 lncRNA activation and miR195 inhibition were observed in both cell types. These significant changes specify and suggest that these regulatory changes may involve malignant neoplasms at the molecular level.

Discussion:

Cervical cancer is a common disease in women, experiments and suggested that it can be caused by HPV infection. It has been observed that lncRNA SNHG1 is activated in tissues and cells of CC, and inhibition of lncRNA SNHG1 prevents cell proliferation, migration and initiates apoptosis. In addition, SNHG1 lncRNA enhances NEK2 expression by ablating miR195 in CC cells.